PLC STM32F407VET6 + W5500 : Modbus TCP (LAN), nap code qua USB-C (CH340C)

Nguon 24VDC

File: power.kicad_sch

MCU STM32F407VET6

File: mcu.kicad_sch

USB nap code

File: usb.kicad_sch

Ethernet

File: comm.kicad_sch

16 ngo ra transistor

File: outputs.kicad_sch

10 ngo vao analog

File: analog.kicad_sch

TỔNG QUAN HỆ THỐNG – PLC STM32F407 + W5500 (Modbus TCP)

- Nguồn 24VDC → bảo vệ (PTC, chống ngược, TVS) + lọc nhiễu → buck 5V → LDO 3.3V; +3.3VA riêng cho ADC.
- MCU STM32F407VET6 (168MHz): 16 ngõ ra DO1–DO16, 10 ngõ vào 4–20mA (ADC 12-bit, VREF 3.000V), SPI2 → W5500.
- Ethernet: module W5500 Lite (có sẵn RJ45): Modbus TCP cổng 502, tối đa 8 kết nối. Coil 00001–00016 = DO1–DO16; Input Register 30001–30010 = AI1–AI10.
- USB-C + CH340C: nạp chương trình qua bootloader UART (tự vào boot bằng DTR/RTS), cổng debug.
- 16 ngõ ra: opto PC817 (cách ly 5kV) + MOSFET 60V/2.5A, kiểu sink, nguồn tải V+/COM riêng 12–30V.
- 10 ngõ vào 4–20mA: TVS + shunt 120Ω + lọc RC; J703 cấp 24V (PTC 0.5A) nuôi cảm biến 2 dây.
- Ghi chú chi tiết (tác dụng, cách tính giá trị) đặt ngay cạnh từng linh kiện ở các trang con.

Sheet: Root

PLC STM32F407 + W5500 Modbus TCP

Sheet: /

File: PLC_STM32_Modbus.kicad_sch

Title: PLC STM32F407 Modbus TCP – 16DO / 10AI

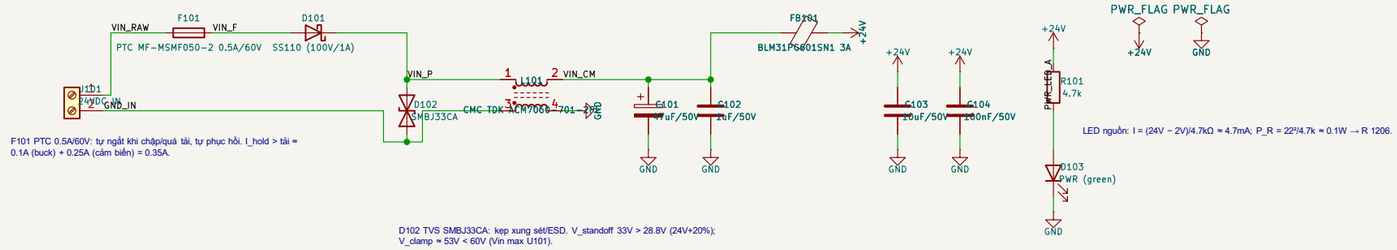
Size: A3	Date: 2026-10-14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/7

NGUON VAO 24VDC: cau chi -> chong nguoc -> TVS -> loc common-mode -> loc LC -> +24V

D101 SS110 (Schottky 100V): chống cắp ngược cực. Sụt ≈ 0.5V → P ≈ 0.5V × 0.35A ≈ 0.18W.

KHỐI VÀO 24VDC – BẢO VỆ & LỌC NHIỄU
Cầu chì → chống ngược → TVS → lọc common-mode → lọc LC → +24V cho toàn board.

FB101 + C103 10µF + C104 100nF: lọc LC thông thấp, chặn nhiễu 2.1MHz của buck đội ngược ra dây nguồn.



F101 PTC 0.5A/60V: tự ngắt khi chập/quá tải, tự phục hồi. I_hold > tải ≈ 0.1A (buck) + 0.25A (cảm biến) = 0.35A.

D102 TVS SMBJ33CA: kẹp xung sét/ESD. V_standoff 33V > 28.8V (24V+20%); V_clamp ≈ 53V < 60V (Vin max U101).

LED nguồn: I = (24V − 2V)/4.7kΩ ≈ 4.7mA; P_R = 22²/4.7k ≈ 0.1W → R 1206.

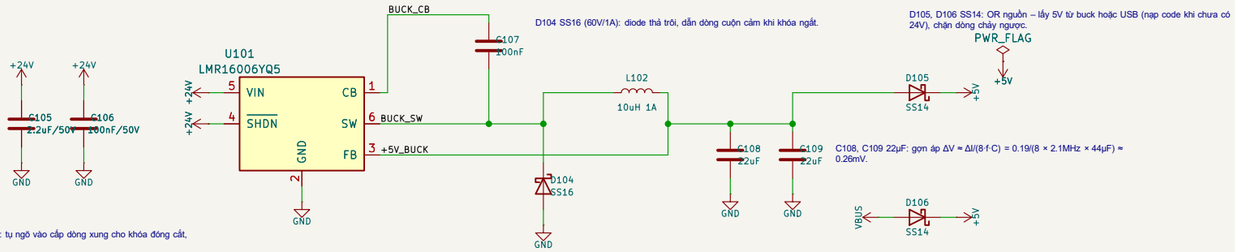
L101 cuộn lọc common-mode: chặn nhiễu đồng pha trên cả 2 dây nguồn (cáp dài, biển tần).

BUCK 24V -> 5V (LMR16006Y, 60V, 0.6A, 2.1MHz)

C101 47µF tụ hóa: dự trữ; ESR dập dao động LC khi cắm nóng. C102 1µF gốm: lọc cao tần.

D104 SS16 (60V/1A): diode thả trôi, dẫn dòng cuộn cảm khi khóa ngắt.

D105, D106 SS14: OR nguồn – lấy 5V từ buck hoặc USB (nạp code khi chưa có 24V), chặn dòng chảy ngược.



C105 2.2µF/50V + C106 100nF: tụ ngõ vào cấp dòng xung cho khóa đóng cắt, đặt sát VIN–GND.

C108, C109 22µF: gợn áp ΔV ≈ ΔI/(8·f·C) = 0.19/(8 × 2.1MHz × 44µF) ≈ 0.26mV.

Cap nguon tu USB khi chua co 24V (de nap code)

BUCK 24V → 5V (LMR16006Y, 0.6A, f_sw = 2.1MHz)
Hạ áp hiệu suất ~85%; D = Vout/Vin = 5/24 ≈ 0.21.

~SHDN nối VIN → luôn chạy. FB nối thẳng ngõ ra (bản 5V cố định, không cần cầu chia).

L102 10µH: ΔI = (Vin−Vout)·D/(L·f) = 19V × 0.21/(10µH × 2.1MHz) ≈ 0.19A (≈ 30% I_out). I_sat ≥ 0.6 + ΔI/2 ≈ 0.7A.

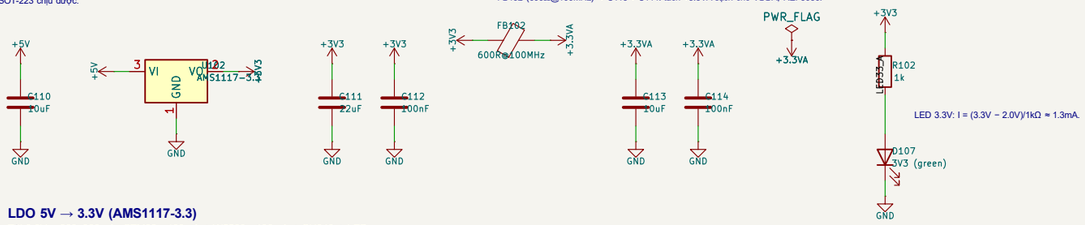
LDO 5V -> 3.3V, loc rieng 3.3VA cho ADC

C110 10µF vào; C111 22µF ra (AMS1117 cần ≥ 22µF để ổn định); C112 100nF lọc cao tần.

C107 100nF: tụ bootstrap lái MOSFET phía cao (datasheet).

Vin ≈ 4.65V ≥ 3.3V + 1.1V (dropout). P = (4.65 − 3.3) × 0.3A ≈ 0.4W → SOT-223 chịu được.

FB102 (600Ω@100MHz) + C113 + C114: tách +3.3VA sạch cho VDDA, REF3030.



LED 3.3V: I = (3.3V − 2.0V)/1kΩ ≈ 1.3mA.

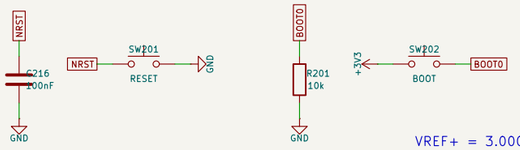
LDO 5V → 3.3V (AMS1117-3.3)
Tải 3.3V ≈ 260–300mA: STM32 ~100mA + W5500 ~130mA + CH340 + LED.

Sheet: Nguon 24VDC
PLC STM32F407 + W5500 Modbus TCP
Sheet: /Nguon 24VDC/
File: power.kicad_sch
Title: Nguon: loc 24VDC, buck 5V, LDO 3.3V, ref 3.0V
Size: A3 | Date: 2026-10-14 | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 2/7

NRST: C216 100nF lọc nhiễu + nút RESET (trở kéo lên 40k có sẵn trong chip).

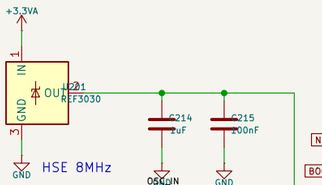
RESET / BOOT

BOOT0: R201 10k kéo xuống → chạy từ Flash. Giữ BOOT khi reset → BOOT0 = 1 → bootloader UART.

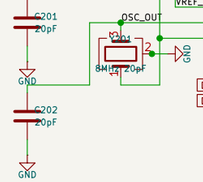


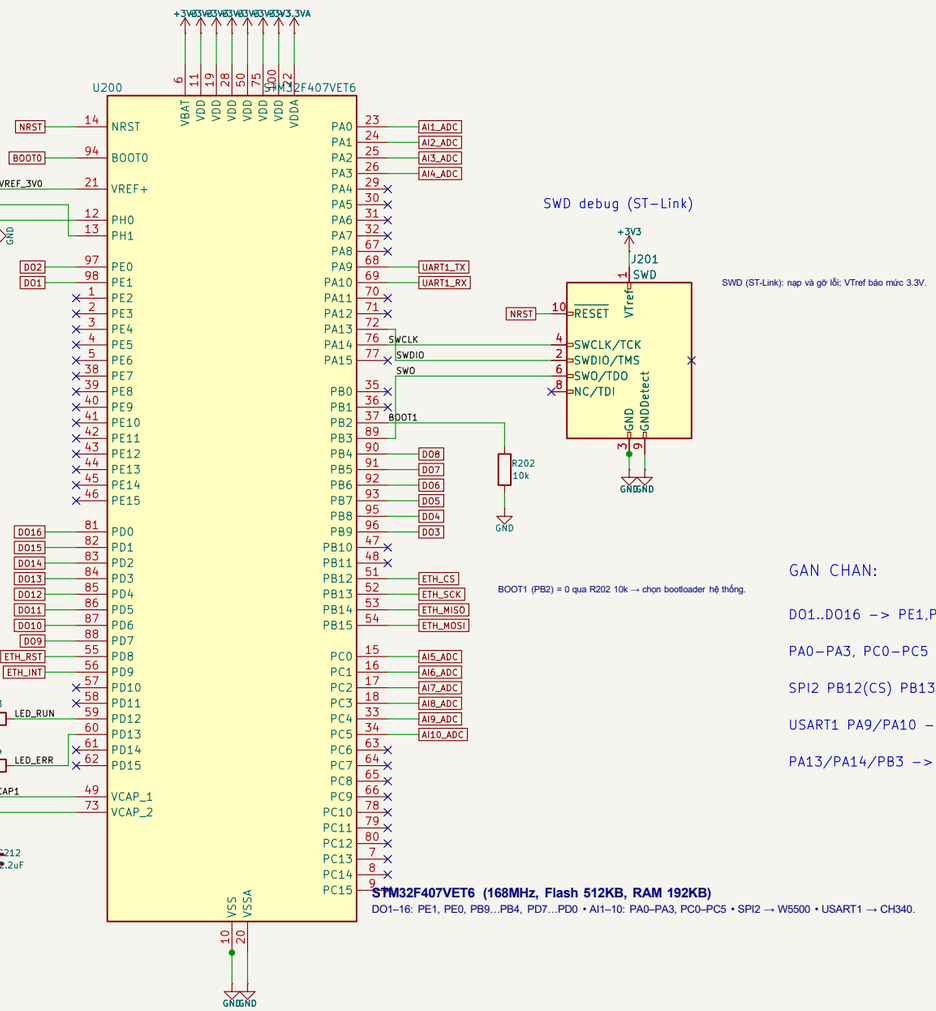
VREF+ = 3.000V

REF3030: VREF+ = 3.000V ±0.2% → 1 LSB = 3.0V/4096 ≈ 0.73mV. C214 1µF + C215 100nF giữ ổn định.



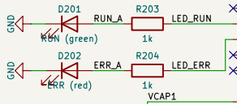
Y201 8MHz → PLL 168MHz. Tụ tải C = 2·(C_L − C_stray) = 2 × (12 − 2) = 20pF (C_L theo datasheet thạch anh).



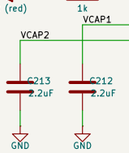


LED trang thai

LED RUN/ERR: I = (3.3V − 2.0V)/1kΩ ≈ 1.3mA (GPIO ≤ 25mA).



VCAP 2.2µF ESR thấp: bắt buộc cho ổn áp 1.2V của lõi (datasheet).



BOOT1 (PB2) = 0 qua R202 10k → chọn bootloader hệ thống.

SWD debug (ST-Link)

SWD (ST-Link): nạp và gỡ lỗi; VTref báo mức 3.3V.

GAN CHAN:

DO1..DO16 -> PE1,PE0,PB9,PB8,PB7,PB6,PB5,PB4,PD7..PD0

PA0–PA3, PC0–PC5 -> AI1..AI10 (ADC1 12-bit)

SPI2 PB12(CS) PB13 PB14 PB15 -> W5500, PD8 RST, PD9 INT

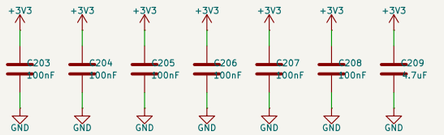
USART1 PA9/PA10 -> CH340C (bootloader UART)

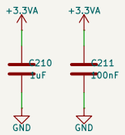
PA13/PA14/PB3 -> SWD, PD12/PD13 -> LED

STM32F407VET6 (168MHz, Flash 512KB, RAM 192KB)

DO1–16: PE1, PE0, PB9…PB4, PD7…PD0 • AI1–10: PA0–PA3, PC0–PC5 • SPI2 → W5500 • USART1 → CH340.

Tu loc nguon: 1 tu 100nF sat moi chan VDD; 1uF+100nF cho VDDA





VDDA: C210 1µF + C211 100nF – nguồn sạch cho ADC.

C203–C208 100nF: một tụ cho mỗi chân VDD, đặt sát chân trên PCB. C209 4.7µF: dự trữ chung (ST AN4488).

Sheet: MCU STM32F407VET6

PLC STM32F407 + W5500 Modbus TCP

Sheet: /MCU STM32F407VET6/

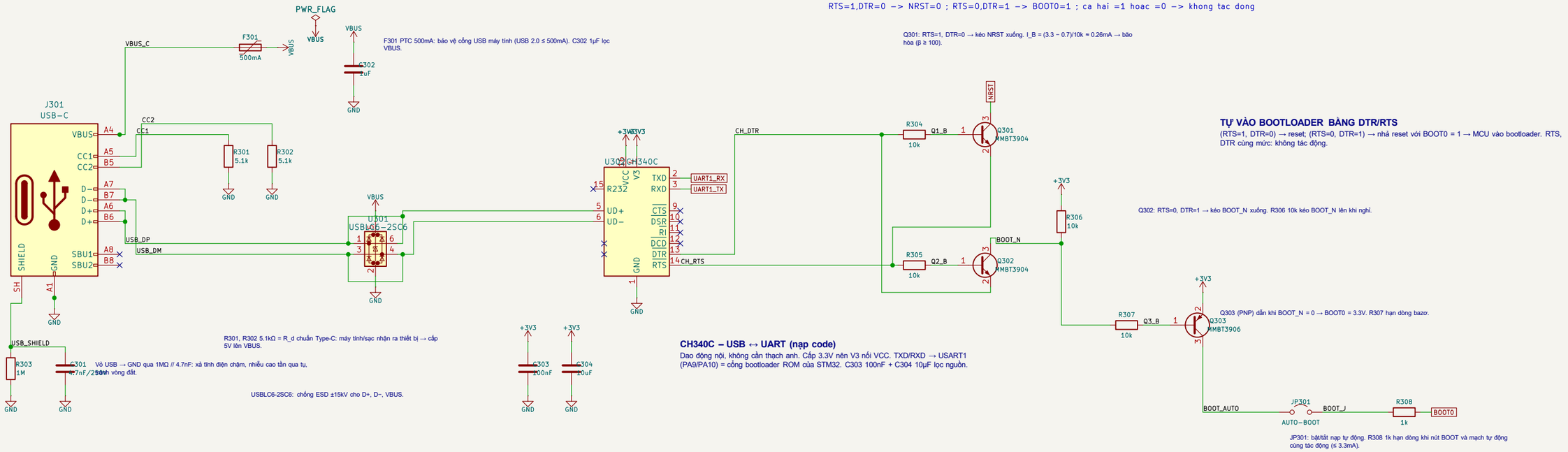
File: mcu.kicad_sch

Title: Vi dieu khien STM32F407VET6

Size: A3	Date: 2026-10-14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 3/7

MACH TU DONG NAP (DTR/RTS) – giong ESP32

RTS=1,DTR=0 –> NRST=0 ; RTS=0,DTR=1 –> BOOT0=1 ; ca hai =1 hoac =0 –> khong tac dong



CH340C – USB ↔ UART (nạp code)
Dao động nội, không cần thạch anh. Cấp 3.3V nên V3 nối VCC. TXD/RXD → USART1 (PA9/PA10) = cổng bootloader ROM của STM32. C303 100nF + C304 10µF lọc nguồn.

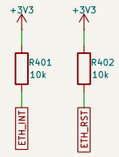
TỰ VÀO BOOTLOADER BẰNG DTR/RTS
(RTS=1, DTR=0) → reset; (RTS=0, DTR=1) → nhả reset với BOOT0 = 1 → MCU vào bootloader. RTS, DTR cùng mức: không tác động.

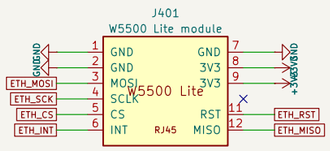
Sheet: USB nap code
PLC STM32F407 + W5500 Modbus TCP
Sheet: /USB nap code/
File: usb.kicad_sch
Title: USB–C –> CH340C –> USART1 + mach tu dong vao bootloader

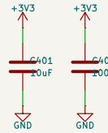
Size: A3	Date: 2026–10–14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 4/7

ETHERNET: module W5500 Lite (W5500 + RJ45 co bien ap) cam vao 2 hang socket 1x6

R401, R402 10k kéo lên INT, RST: giữ mức 1 khi MCU chưa khởi động (module không reset ngẫu nhiên). INT tích cực mức thấp.







C401 10µF + C402 100nF: lọc nguồn 3.3V cho module, đặt sát chân 3V3 của socket. W5500 hút ~130mA dạng xung khi phát → cần tụ dự trữ gần module.

MODULE W5500 LITE – ETHERNET 10/100 + TCP/IP PHẦN CỨNG

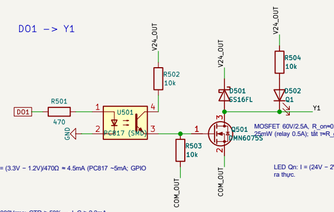
Trên module có sẵn: chip W5500, thạch anh 25MHz, điện trở phối hợp 49.9Ω, tụ TOCAP/1V2O và jack RJ45 có biến áp cách ly – board chính chỉ cần cấp 3.3V và nối SPI.
8 socket phần cứng = tối đa 8 kết nối Modbus TCP đồng thời (cổng 502).
Giao tiếp MCU: SPI2 – PB12 = CS, PB13 = SCLK, PB14 = MISO, PB15 = MOSI (≤ 21MHz), PD8 = RST, PD9 = INT.
Lắp: 2 hàng socket cái 1×6 bước 2.54mm, cách nhau 20.32mm; RJ45 hướng ra mép board. Hàn header vào module với chân hướng xuống (linh kiện và RJ45 ở mặt trên).

Sheet: Ethernet
PLC STM32F407 + W5500 Modbus TCP
Sheet: /Ethernet/
File: comm.kicad_sch
Title: Ethernet – module W5500 Lite (Modbus TCP)
Size: A3 | Date: 2026-10-14 | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 5/7

16 NGO RA TRANSISTOR SINK (NPN–type): MCU –> PC817 –> MOSFET 60V/2.5A. Tai noi giua V+ (24V) va Yn. Cap nguon ngoai vao V+/COM (cach ly voi mach logic).

R_c ≤ 10k + R_g 10k: V_GS = V+ × 10k/20k = V+/2 ≈ 12V @24V
(V_GS max 20V → V+ ≤ 30V; ≥ 4.5V → V+ ≥ 10V). I_opto =
24V/20k = 1.2mA < 2.2mA → bão hòa.

DO1 –> Y1



R_in 470Ω: I_F = (3.3V − 1.2V)/470Ω ≈ 4.5mA (PC817 ~5mA; GPIO ≤ 25mA).

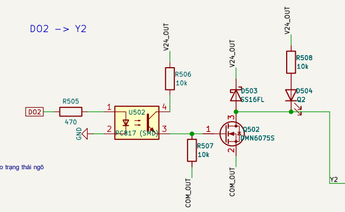
MOSFET 60V/2.5A, R_on=0.1Ω: P = 0,5²×0.1 = 25mW (relay 0.5A); tắt =tR_g·C_iss=3µs.

LED Qn: I = (24V − 2V)/10kΩ ≈ 2.2mA – báo trạng thái ngõ ra thực.

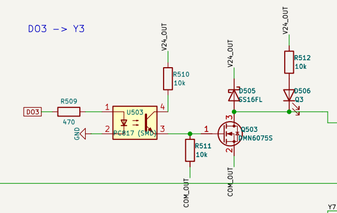
PC817: cách ly 5000Vrms; CTR ≥ 50% → I_C ≥ 2.2mA.

SS16: đập xung ngược L·di/dt của cuộn relay về V+.

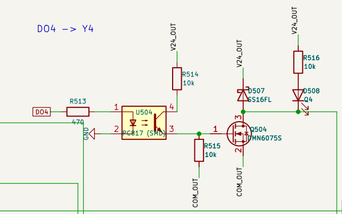
DO2 –> Y2



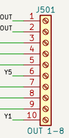
DO3 –> Y3

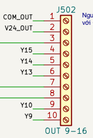


DO4 –> Y4



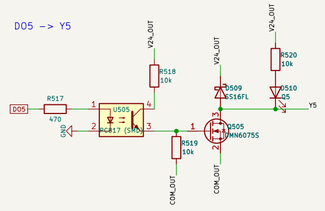
CAU DAU NGO RA



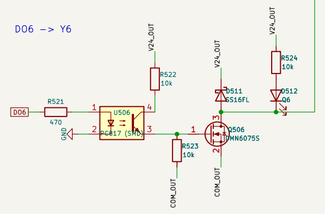


Nguồn relay 12–30V vào V+ (chân 2) / COM (chân 1). Tải nối giữa V+ và Yn. KHÔNG nối COM với GND board (mất cách ly). Giới hạn 30V: Vceo PC817 = 35V.

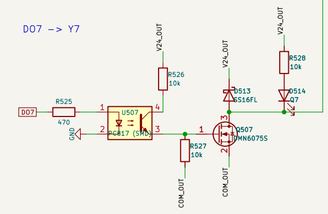
DO5 –> Y5



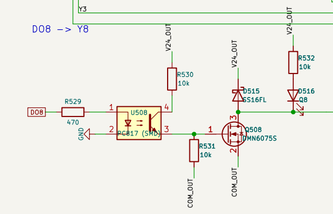
DO6 –> Y6



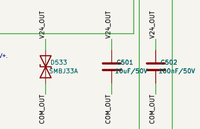
DO7 –> Y7



DO8 –> Y8

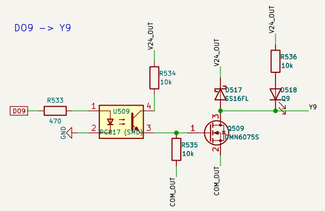


Loc / bao ve nguon ngo ra

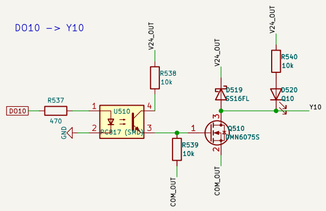


D533 TVS 33V + C501 10µF/50V + C502 100nF: bảo vệ và lọc nguồn tải V+.

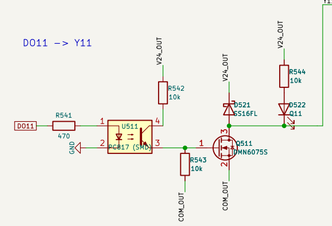
DO9 –> Y9



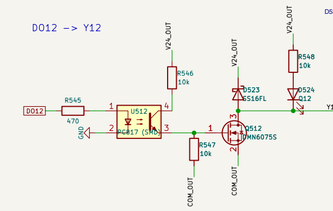
DO10 –> Y10



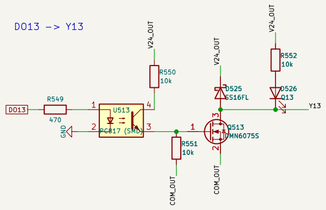
DO11 –> Y11



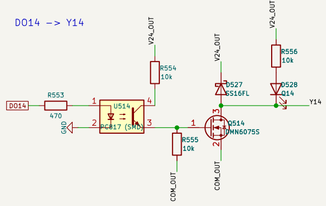
DO12 –> Y12



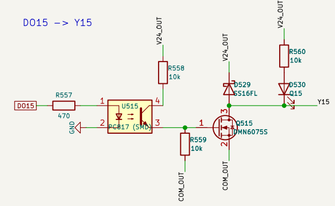
DO13 –> Y13



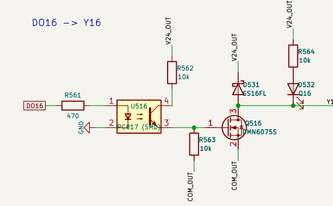
DO14 –> Y14



DO15 –> Y15



DO16 –> Y16



KÊNH NGÕ RA (16 kênh giống nhau) – SINK, CÁCH LY QUANG

DOx = 1 → LED opto sáng → MOSFET dẫn → Yn kéo về COM → relay hút.

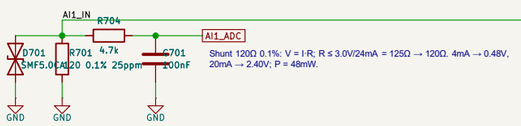
Sheet: 16 ngo ra transistor	
PLC STM32F407 + W5500 Modbus TCP	
Sheet: /16 ngo ra transistor/	
File: outputs.kicad_sch	
Title: 16 ngo ra transistor (sink), cach ly quang	
Size: A2	Date: 2026-10-14
KiCad E.D.A. 10.0.6	**Rev: 1.0**
	Id: 6/7

10 NGO VAO 4–20mA: shunt 120R 0.1% –> Vadc = I x 120R ; 4mA –> 0.48V ; 20mA –> 2.40V ; 24mA –> 2.88V (VREF = 3.000V)

TVS SMF5.0CA: kẹp xung/quá áp > 5V. Không có cầu chì: đấu nhầm 24V vào AIn → 200mA, 4.8W → cháy shunt.

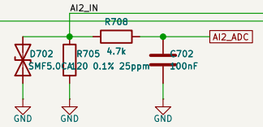
AI1: 4–20mA

RC 4.7k/100nF: f_c = 1/(2πRC) ≈ 340Hz; hạn dòng vào ADC ≈ 1mA khi quá áp.



Shunt 120Ω 0.1%: V = I·R; R ≤ 3.0V/24mA = 125Ω → 120Ω. 4mA → 0.48V, 20mA → 2.40V; P = 48mW.

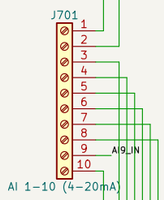
AI2: 4–20mA



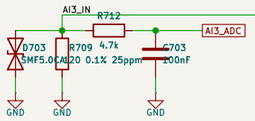
KÊNH 4–20mA (10 kênh giống nhau)

Firmware: I(mA) = raw/4095 × 3000mV/120Ω. Dải 4–20mA ≈ 2600 mức ADC.

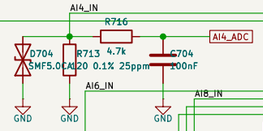
J701: AI1–AI10 nối thẳng vào kênh (đã bỏ cầu chì từng kênh).



AI3: 4–20mA



AI4: 4–20mA



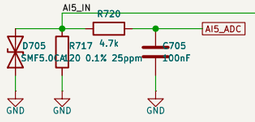
AI6_IN

AI8_IN

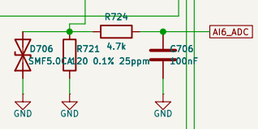
AI10_IN

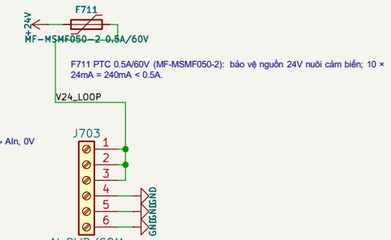
Nguon cap cho cam bien 2 day (24V loop) + chan chung GND

AI5: 4–20mA



AI6: 4–20mA

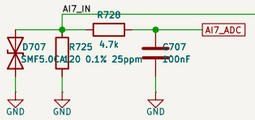




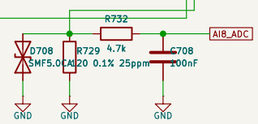
F711 PTC 0.5A/60V (MF-MSMF050-2): bảo vệ nguồn 24V nuôi cảm biến; 10 × 24mA = 240mA < 0.5A.

Cảm biến 2 dây: V24_LOOP → (+) → (–) → AIn. 4 dây: ngõ ra mA → AIn, 0V cảm biến → GND.

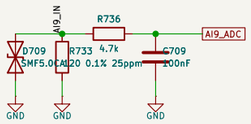
AI7: 4–20mA



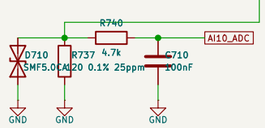
AI8: 4–20mA



AI9: 4–20mA



AI10: 4–20mA



Sheet: 10 ngo vao analog

PLC STM32F407 + W5500 Modbus TCP

Sheet: /10 ngo vao analog/

File: analog.kicad_sch

Title: 10 ngo vao analog 4–20mA

Size: A3	Date: 2026–10–14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 7/7